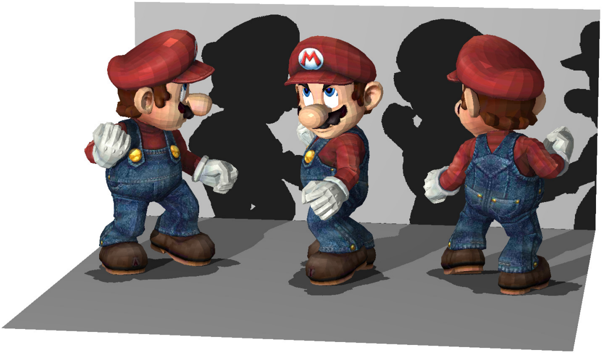
Computer Graphics



Benny Van Houdt

Universiteit Antwerpen

Computer Graphics Doelstellingen

Theorie

→ Inzicht verkrijgen in volgende zaken:

- De algoritmen die gebruikt worden in 3D-engine.
- De vergelijkingen die in deze algoritmen voorkomen.
- Waar komen deze vergelijkingen vandaan?

Practica

→ Aanscherpen van programmeervaardigheden door stapgewijs een eigen 3D-engine te implementeren *from scratch* in C++ (of python voor WIS)

Computer Graphics Onderdelen

Theorie

- DO 10u45-12u45, lokaal A143
- Offline alternatief: Videos op BB (Studiemateriaal)
- Uitgewerkte cursus beschikbaar (86 pagina's)
- Te koop via cursusdienst en pdf op BB

Practica: 2 Groepen

- Groep 1: WOE 13u45-16u45 (G.025)
- Groep 2: VR 13u45-16u45 (G.025)

Computer Graphics Examenvorm

Theorie

- Schriftelijk examen: 10 punten (of minder)
- Examens van voorgaande jaren op BB
- Gesloten boek. Eén of meer vragen per hoofdstuk

Practica

- 10-tal opdrachten
- 6 punten t/m Z-buffering met driehoeken = MUST DO
- max 12 punten
- Geen verplichte tussentijdse evaluatie (wel aangeraden)

Eindscore

- Moet minstens 5/10 scoren op beide onderdelen om te slagen!

Hoofdstukken en opdrachten (red is MUST DO)

- 2D L-Systemen (1,25)
- 3D lijntekening (1)
- 3D lichamen en L-systemen (1)
- Z-Buffering met lijnen (1)
- Z-Buffering met driehoeken (1,5)
- 3D-fractalen (1)
- Clipping (0,5)
- Belichting (1,5)
- Schaduw (1)
- Texture mapping (1,5)
- 3D-lijntekeningen met cylinders (0,75)

Computer Graphics Best Student Award

Wat brengt het op?

- Ingekaderd Certificaat
- Eeuwige vermelding op website:

  win.uantwerpen.be/∼vanhoudt/

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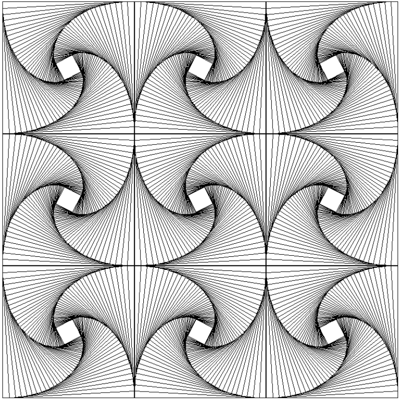
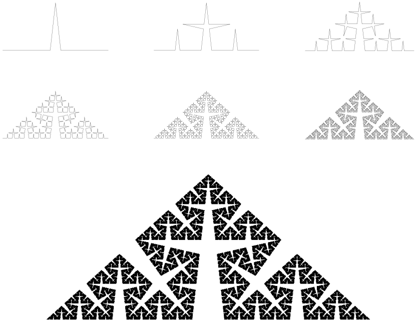
Hoofdstuk 1: 2D-Lijntekeningen





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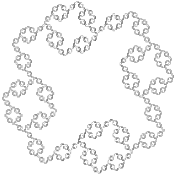
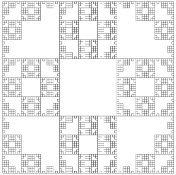
Hoofdstuk 1: L-systemen

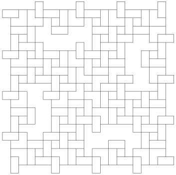
n=4, F $\rightarrow$ FF−F−F−F−F−F+F



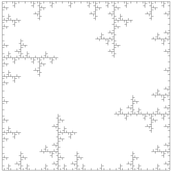
n=4, F $\rightarrow$ FF−F−F−F−FF



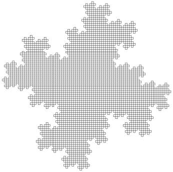
n=3, F $\rightarrow$ FF−F+F−F−FF



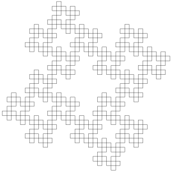
n=4, F $\rightarrow$ FF−F−−F−F



n=5, F $\rightarrow$ F−FF−−F−F

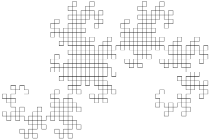


n=4, F $\rightarrow$ F−F+F−F−F

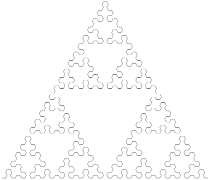


Hoofdstuk 1: L-systemen

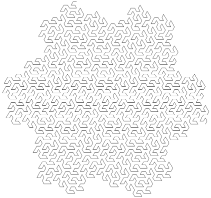
n=10, I=F, $\delta=\pi/2$,
$F \rightarrow F{+}G{+}$, $G \rightarrow -F-G$



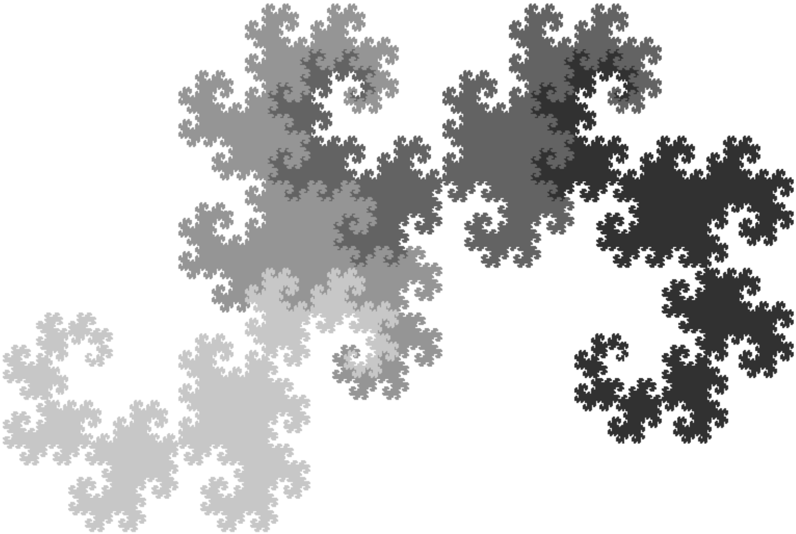
n=6, I=F, $\delta=\pi/3$,
$F \rightarrow G-F-G$, $G \rightarrow F{+}G{+}F$



n=4, I=F, $\delta=\pi/3$,
$F \rightarrow F{+}G{+}{+}G-F--FF-G{+}$,
$G \rightarrow -F{+}GG{+}{+}G{+}F--F-G$

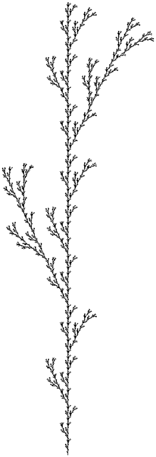


Hoofdstuk 1: L-systemen

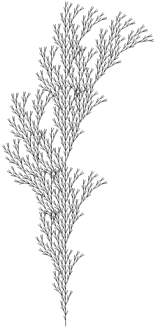


Hoofdstuk 1: L-systemen met haakjes

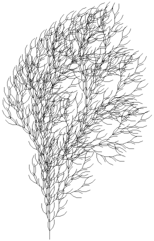
n=5, $\delta=2\pi$ (25.7/360),
F$\rightarrow$ F[+F]F[−F]F



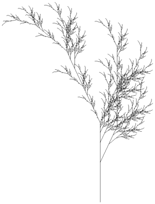
n=5, $\delta=2\pi$ (20/360),
F$\rightarrow$ F[+F]F[−F][F]



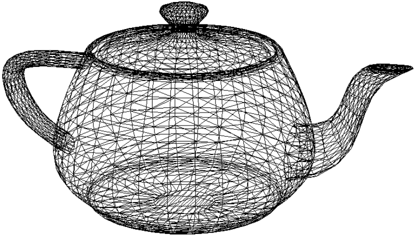
n=4, $\delta=2\pi$ (22.5/360),
F$\rightarrow$ FF−[−F+F+F]+[+F−F−F]



n=6, $\delta=2\pi$ (25/360), I=+X,
F$\rightarrow$ FF, X$\rightarrow$ F−[[X]+X]+F[+FX]−X



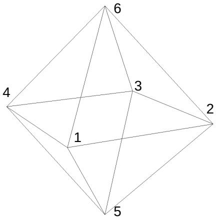
Hoofdstuk 2: Wireframe tekeningen in 3D (zwart-wit)

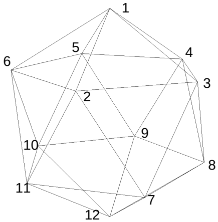


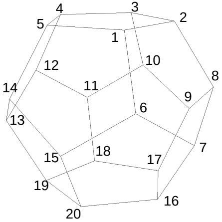
Nodige theorie:

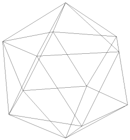
- 2D- en 3D-rotatie matrices
- Transformatie naar Eye-coördinaatsysteem
- Perspectief projectie

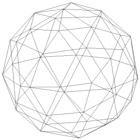
Hoofdstuk 2: Platonische Lichamen en de Bol

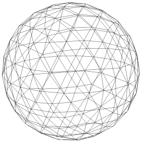


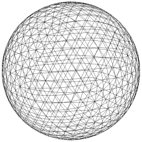




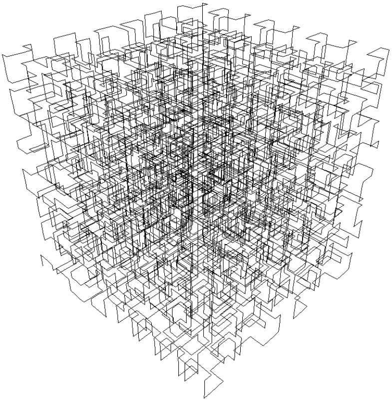


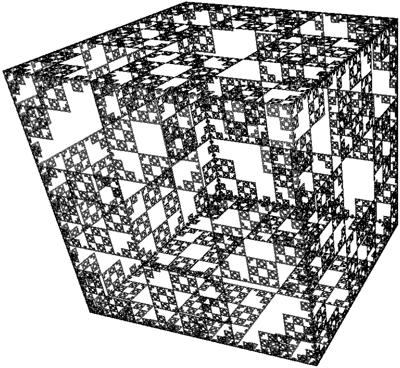




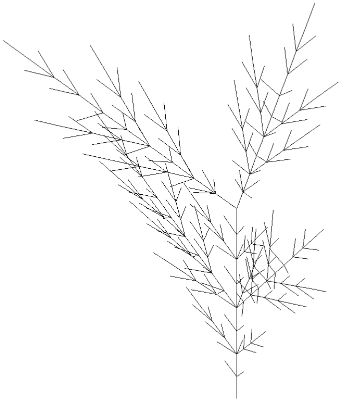


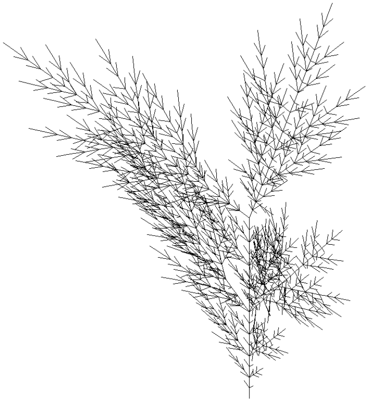
Hoofdstuk 2: 3D L-systemen





Hoofdstuk 2: 3D L-systemen met haakjes





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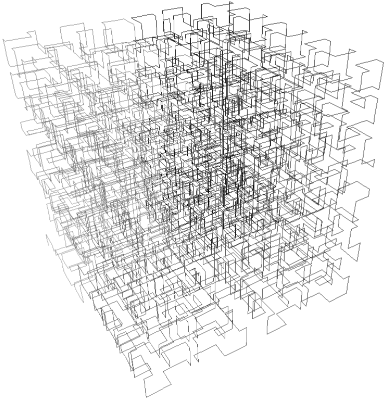
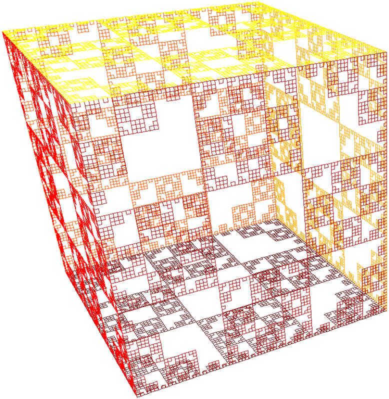
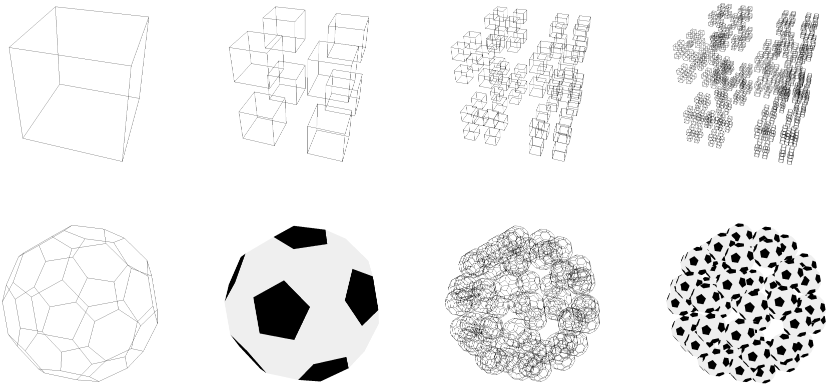
Hoofdstuk 3: Z-buffering, Wireframes in kleur





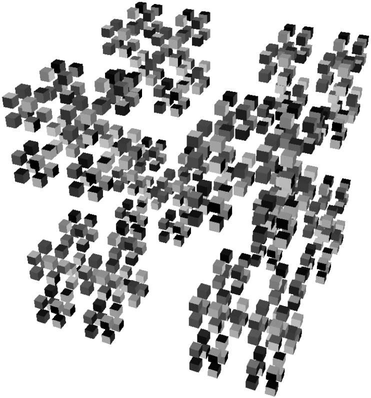
Nodige theorie: Z-buffering met lijnen en $1/z$-interpolatie

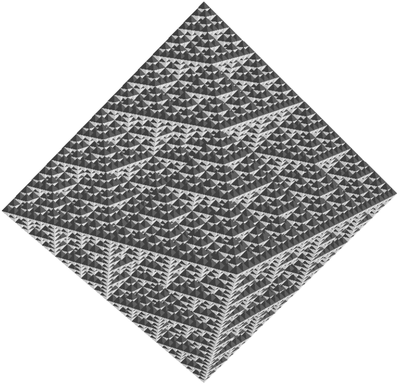
Hoofdstuk 3: Z-buffering, 3D-fractalen



Nodige theorie: Inkleuren van een driehoek en Z-buffering met driehoeken

Hoofdstuk 3: Z-buffering, 3D-fractalen





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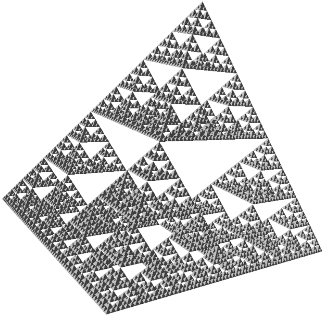
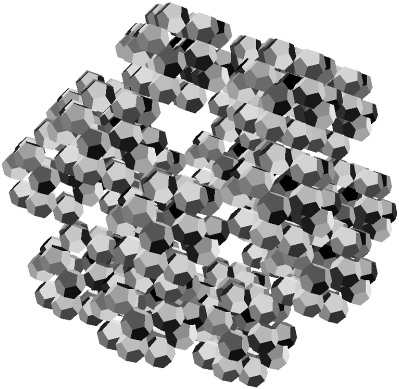
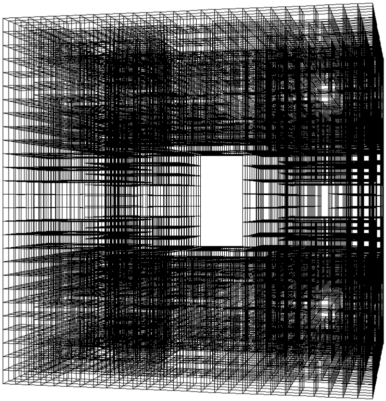
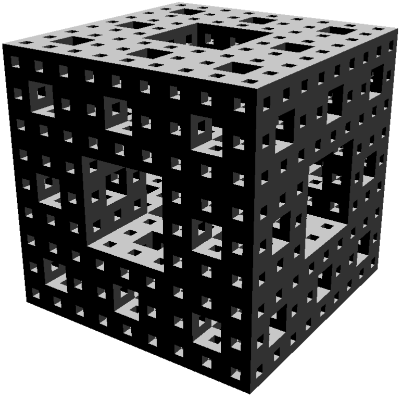
Hoofdstuk 3: Z-buffering, 3D-fractalen





Hoofdstuk 3: Z-buffering, 3D-fractalen





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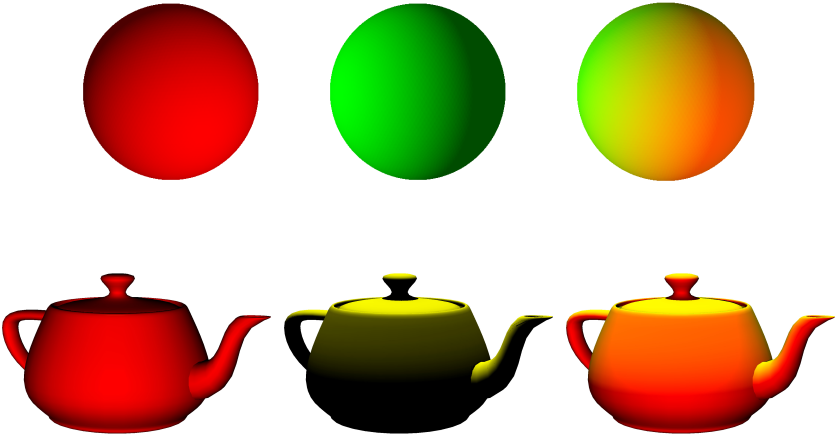
Hoofdstuk 4: Ambient en Diffuus licht (op oneindig)



Nodige theorie: RGB model

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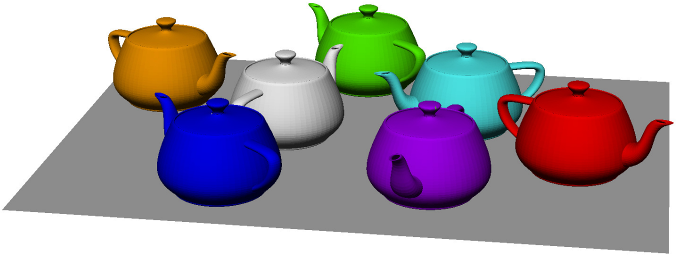
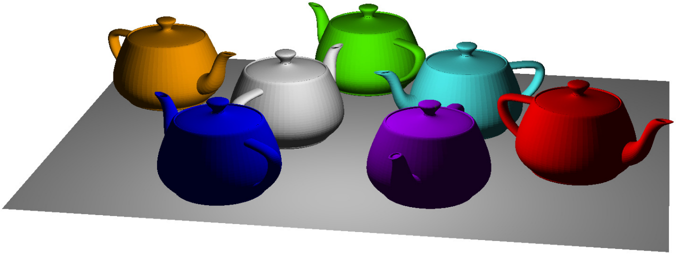
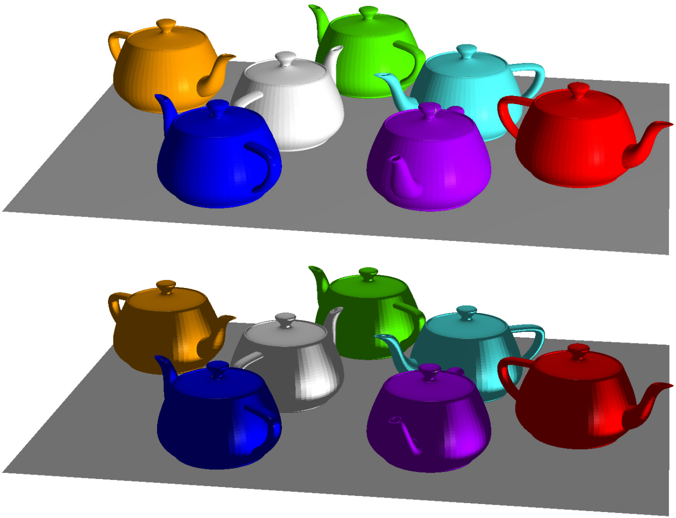
Hoofdstuk 4: Diffuus licht (op oneindig vs. puntbron)





Hoofdstuk 4: Spotlicht en spiegelend licht



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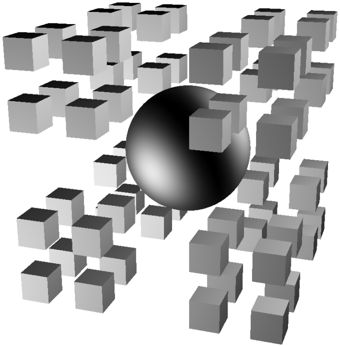
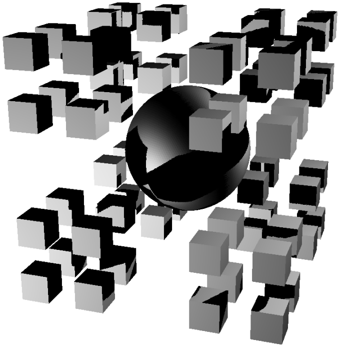
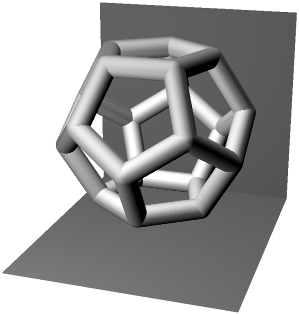
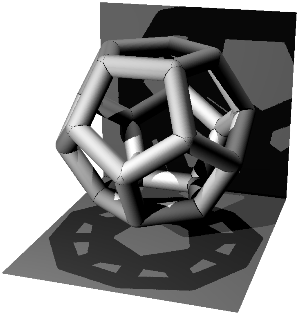
Hoofdstuk 4: Schaduw en Z-buffering





Hoofdstuk 4: Schaduw en Z-buffering





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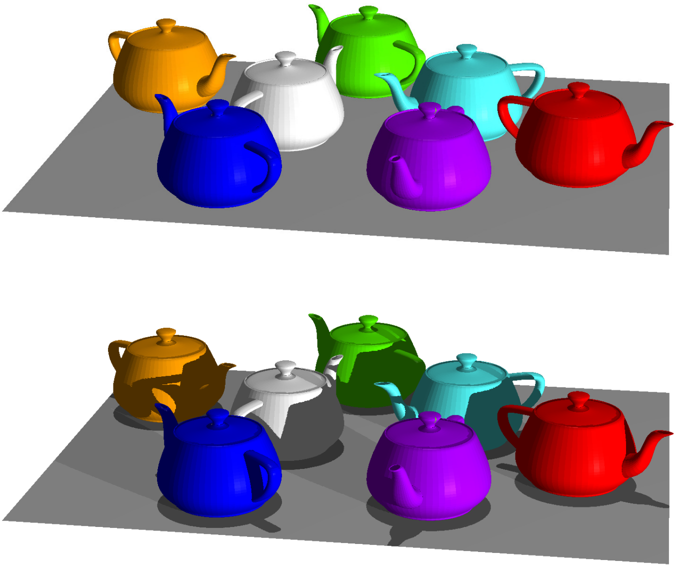
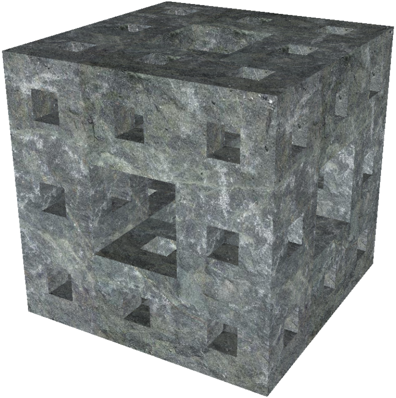
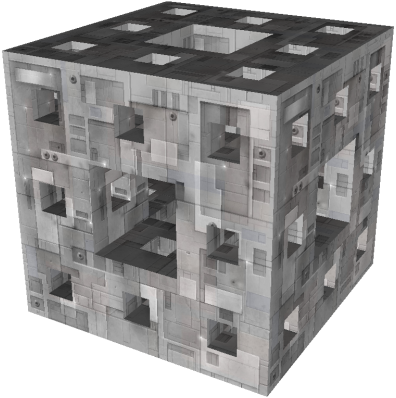
Hoofdstuk 4: Schaduw en Z-buffering

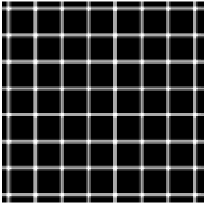
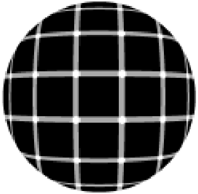
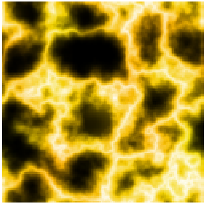
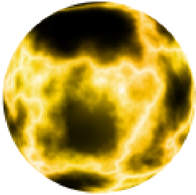


Hoofdstuk 5: Texture Mapping op een vlak





Hoofdstuk 5: Texture Mapping op een bol

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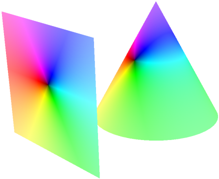
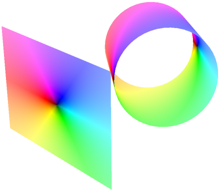
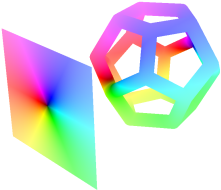
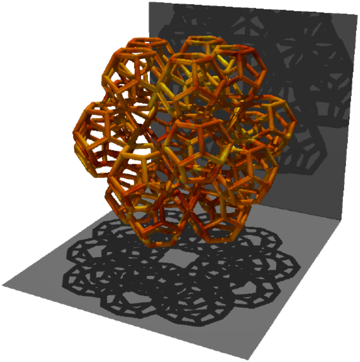
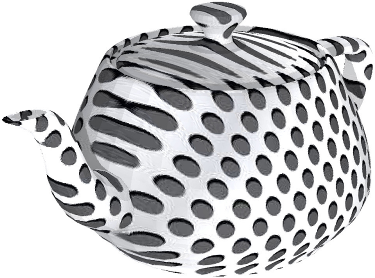
Hoofdstuk 5: Texture Mapping op willekeurig oppervlak





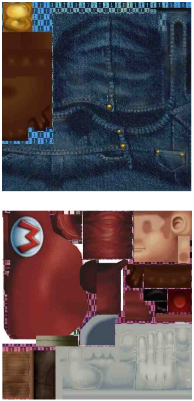






Hoofdstuk 5: Cube mapping (sinds 2020)

Hoofdstuk 5: Textuurcoördinaten (sinds 2020)



Hoofdstuk 5: Input normaalvectoren (sinds 2020)

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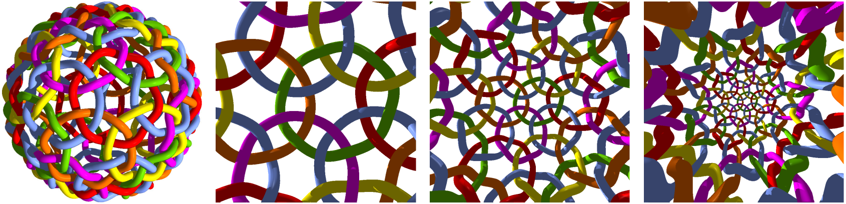
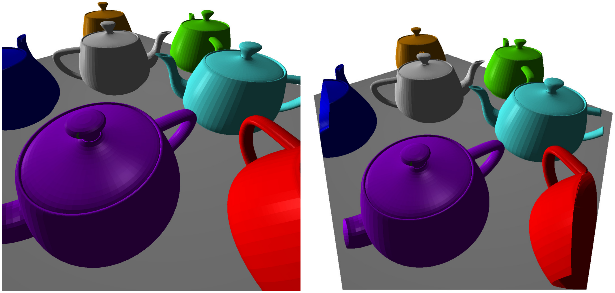
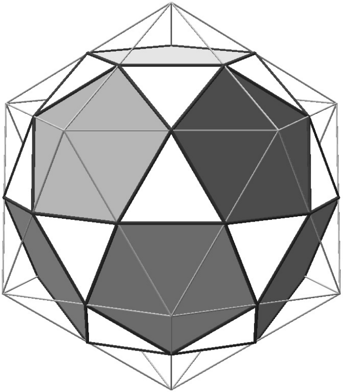
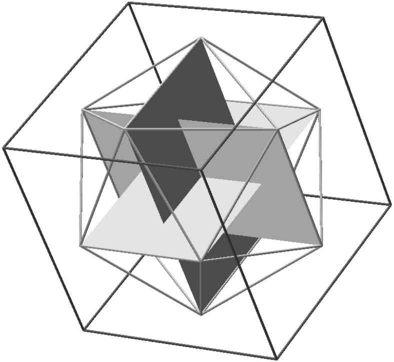
Hoofdstuk 3: Clipping (sinds 2020)





Lichamen (links: Icosidodecahedron)





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